

REST API Authentication — Types

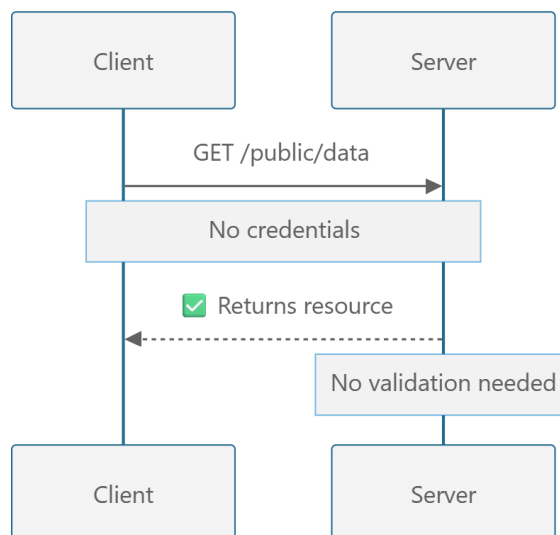
Authentication Types:

1. No Auth
2. Basic Auth
3. API Key
4. Bearer Token (JWT)
5. OAuth 2.0

🚫 1. NO AUTH

No authentication required. Anyone can access the API endpoint.

Workflow



Example Request

GET /public/data HTTP/1.1

Host: api.example.com

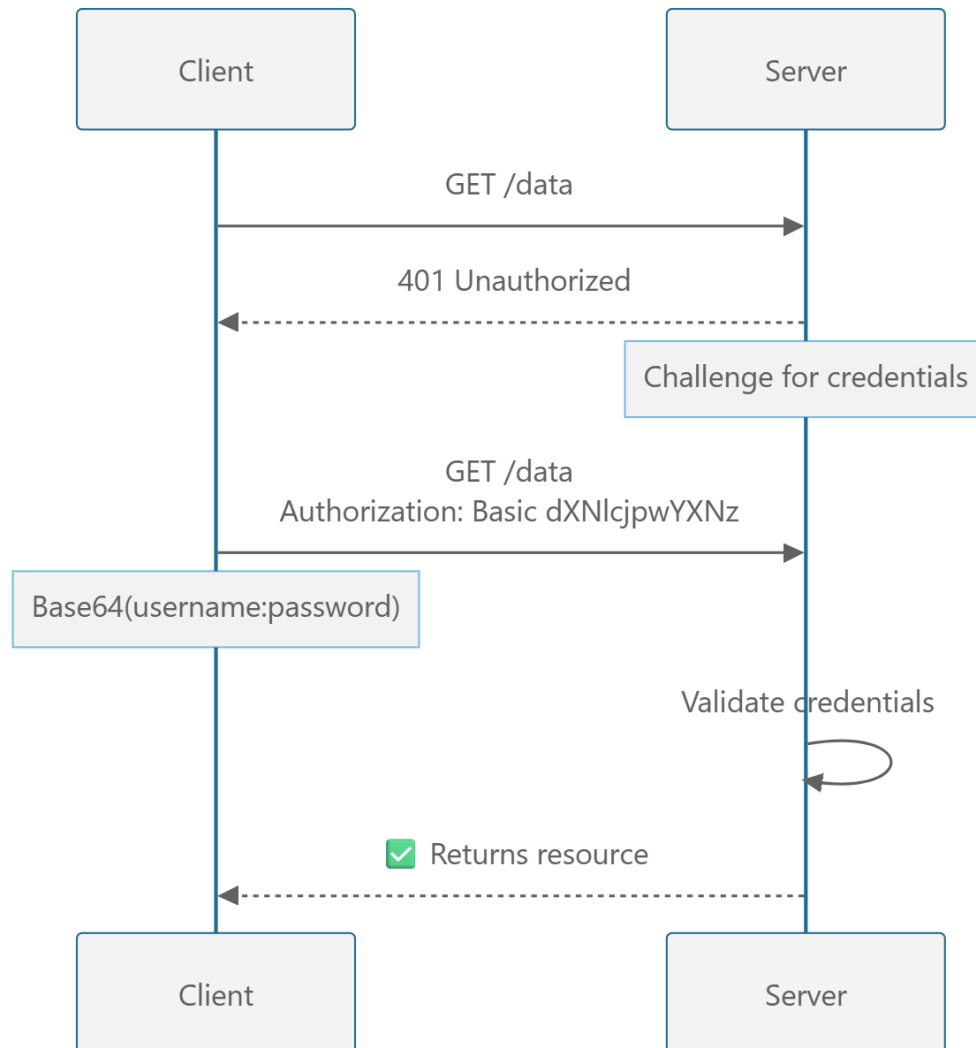
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Use Case	Examples
Public APIs	Weather data, Public holidays
Open Data	Government datasets
Health Checks	Monitoring endpoints

🔒 2. BASIC AUTH

Uses username and password encoded with Base64.

How It Works



Example Request

`GET /data HTTP/1.1`

`Host: api.example.com`

`Authorization: Basic dXNlcjpwYXNz`

⚡ **Tip:** `Base64("username:password") = dXNlcjpwYXNz`

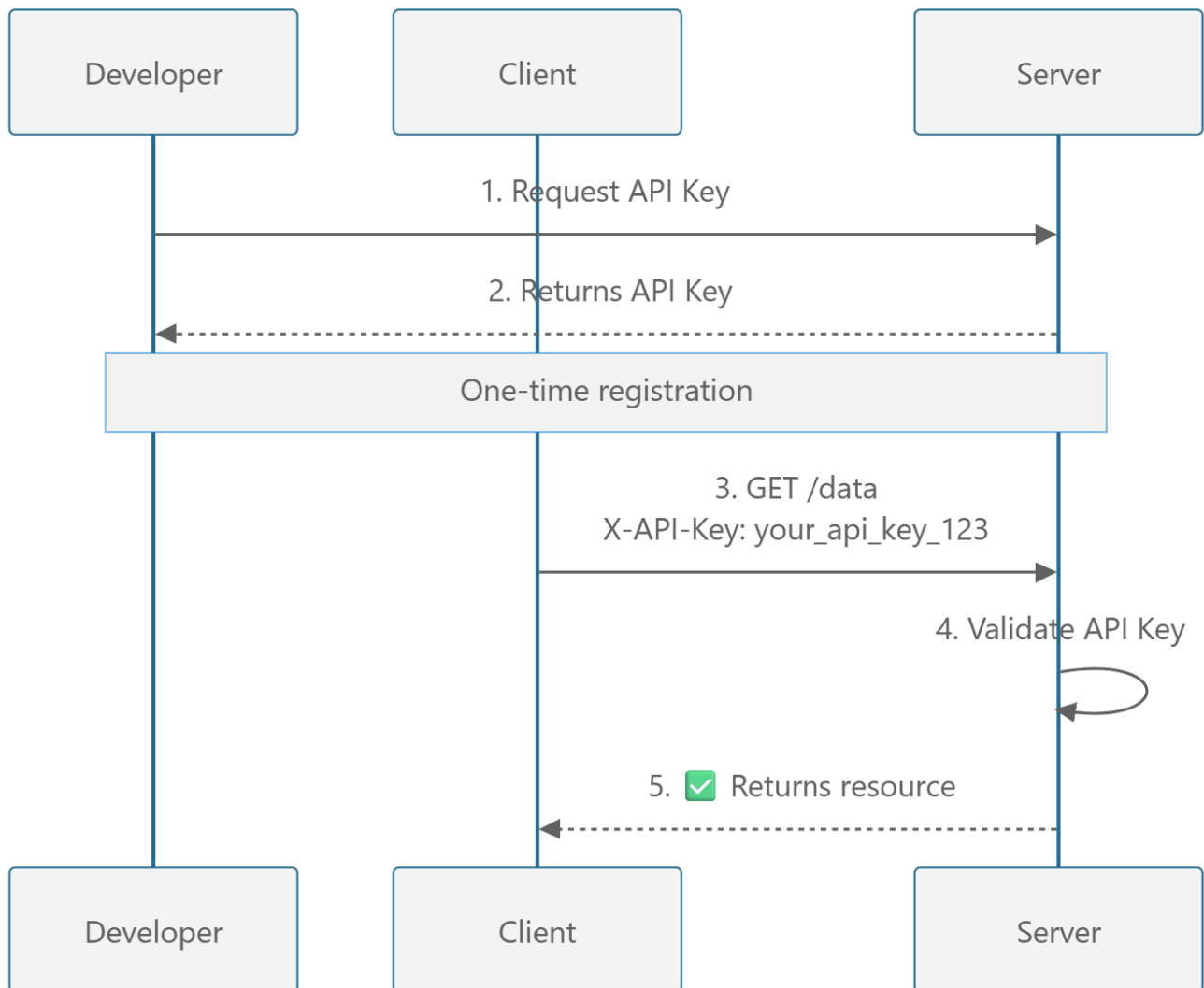
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Use Case	Examples
Simple Internal Systems	Admin panels
Quick Testing	Development environments
Legacy Systems	Older APIs

🔑 3. API KEY

A unique key is sent with each request to identify the client.

Workflow



Example Request

GET /data HTTP/1.1

Host: api.example.com

X-API-Key: your_api_key_123

⚡ **Tip:** API keys are usually sent in headers (X-API-Key) or as query params (?api_key=xxx)

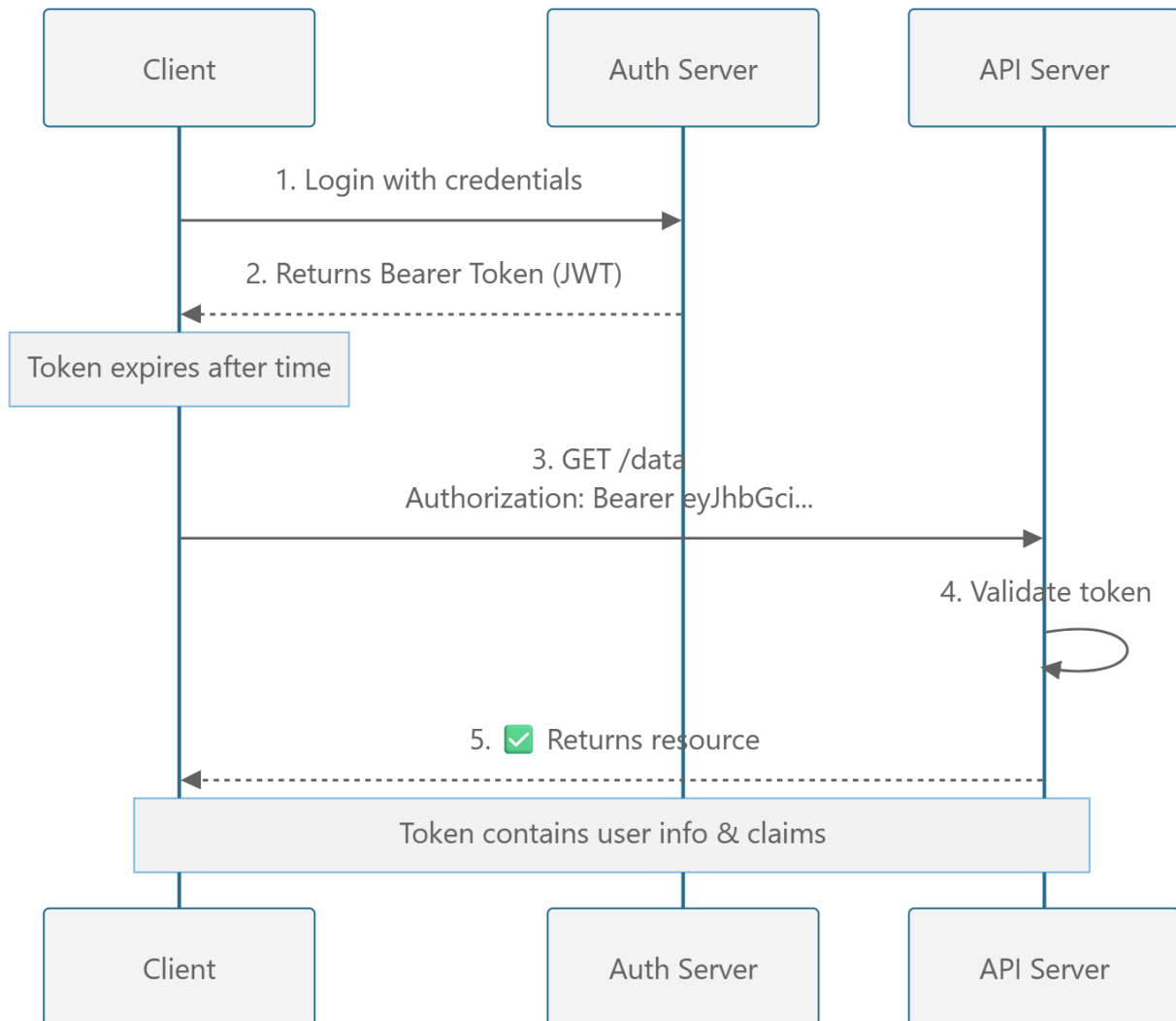
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Use Case	Examples
Public APIs	Weather, News APIs
Dashboards	Analytics services
Rate-Limited Services	Payment gateways

🚩 4. BEARER TOKEN (JWT)

A token (usually JWT) is sent in the Authorization header as Bearer token.

Workflow



Example Request

GET /data HTTP/1.1

Host: api.example.com

Authorization: Bearer eyJhbGciOiJIbGciOiJIbGci...

⚡ **Tip:** Tokens expire. Refresh or re-login when needed.

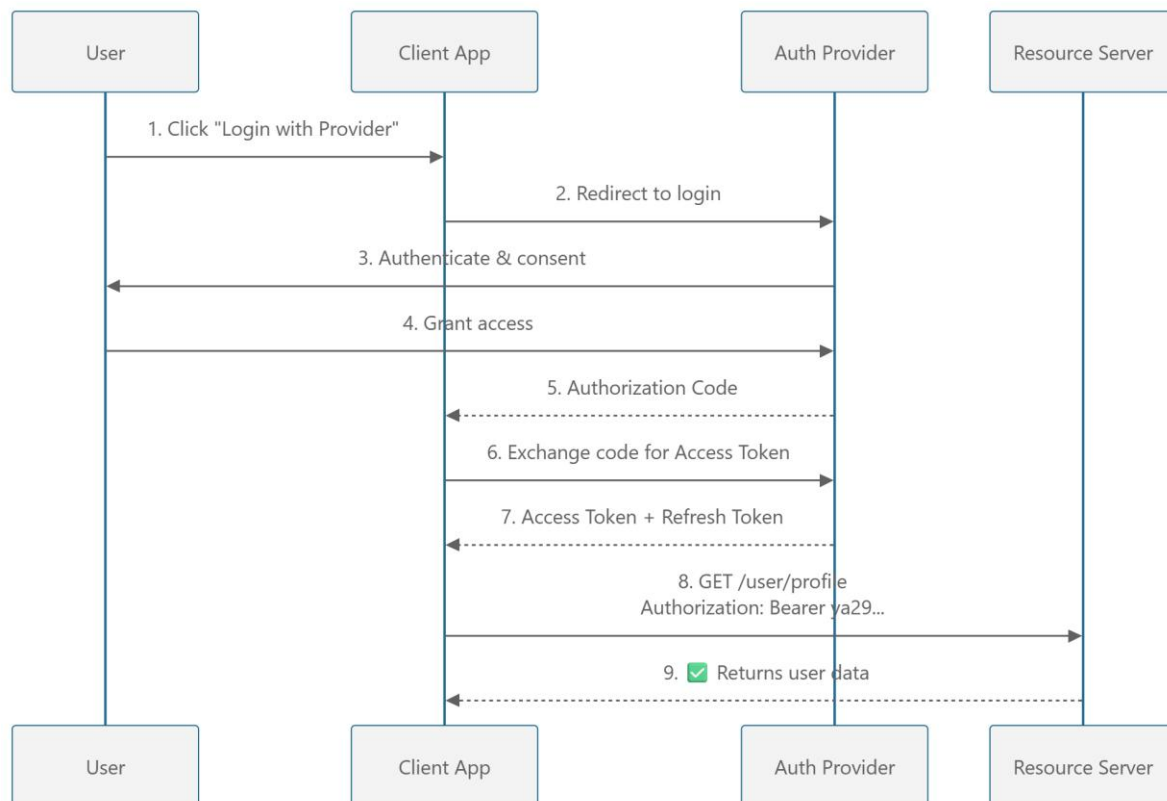
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Use Case	Examples
Mobile Apps	iOS/Android applications
SPAs	React, Vue, Angular apps
Secured APIs	Banking, Healthcare systems

5. OAUTH 2.0

Delegated authorization. Allows access to user data without sharing passwords.

Authorization Code Flow (Most Common)



Example Request

GET /user/profile HTTP/1.1

Host: api.example.com

Authorization: Bearer ya29.a0AfH6SMD...

OAuth 2.0 Flows

Flow	Use Case	Status
Authorization Code	Web & Server Apps	✅ Recommended
Client Credentials	Server-to-Server	✅ Recommended

Best For

Use Case	Examples
Social Login	Google, Facebook, GitHub
Third-party Access	Microsoft, Twitter
Enterprise SSO	Okta, Auth0

Quick Comparison Table

Feature	No Auth	Basic Auth	API Key	Bearer Token	OAuth 2.0
Security Level	✗ None	⚠ Low	🔒 Medium	🔒🔒 High	🔒🔒🔒 Very High
Ease of Implementation	🟢 Very Easy	🟢 Easy	🟡 Medium	🟡 Medium	🔴 Complex
User Login Required	✗ No	✅ Yes	✗ No	✅ Yes	✅ Yes
Token Expiry	N/A	N/A	✗ Usually No	✅ Yes	✅ Yes
Refresh Tokens	N/A	N/A	✗ No	✅ Yes	✅ Yes
Rate Limiting Support	✗ No	✗ No	✅ Yes	✅ Yes	✅ Yes
Best For	Public data	Internal tools	Public APIs	Secured APIs	User data access